# 🌿 BIOLOGY CH.16 — PLANT KINGDOM — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

## 📄 Topics in this chapter:



## 1. Bioremediation — Keeping the Environment Clean

- 'Bacteria aur Fungi' environment ko clean rakhne mein help karte hain (Bioremediation) — Bacteria (Oleispira) oil spills clean karte hain, Fungi mycoremediation mein use hote hain. *[Q866]*

## 2. Tropism — Types of Directional Plant Growth

- ⚠️ Trap: Pollen tubes ka ovules ki taraf growth 'Chemotropism' hai, Geotropism NAHI — Sunflower phototropic hai, roots downward migration positive geotropism hai, Hydrotropism moisture ki taraf growth hai. *[Q867]*
- Plant part light se 'door' grow karta hai to 'Negative Phototropism' kehte hain (roots mein), light ki taraf grow karta hai to Positive Phototropism (stems mein). *[Q979, Q992]*
- Plant organ ka growth stimulus ki taraf 'Positive Tropism' kehlaata hai — roots water/gravity ki taraf, stems light ki taraf. *[Q983]*
- Gravity ki taraf plant part ka movement 'Gravitropism' (Geotropism) kehlaata hai — Phototropism light response, Hydrotropism water response, Chemotropism chemical stimulus response hai. *[Q1003]*

## 3. Stomata & Gas Exchange

- Leaves mein 'Stomata' mein respiration ke dauran massive gas exchange hoti hai — microscopic pores hain. Grana photosynthesis light reactions ka site hai, Thylakoids chloroplast ki internal membranes hain. *[Q869]*
- Guard cells water absorb karke swell hoti hain (turgid banti hain), jisse Stomata open hoti hain — water lose karne par flaccid hokar stomata close ho jaati hain. *[Q871]*
- ⚠️ Trap: 'Intracellular spaces' transpiration ka main site NAHI hai — Stomata, Cuticle aur Lenticels transpiration ke main types (stomatal, cuticular, lenticular) hain. *[Q875]*
- Big plants-trees ke hard stem mein gaseous waste 'Lenticels' se remove hoti hai — porous tissues hain jo woody stems mein gas exchange allow karte hain. *[Q876]*
- Raat mein photosynthesis nahi hoti, isliye 'Carbon dioxide' removal main gas exchange hai — plants din mein $CO_2$ lete hain-$O_2$ dete hain, raat mein $O_2$ lete hain-$CO_2$ dete hain. *[Q877]*
- Din ke time plants mein major event '$O_2$ release' hai — Photosynthesis equation: $6CO_2 + 6H_2O + \text{Sunlight} \rightarrow C_6H_{12}O_6 + 6O_2$. *[Q890]*

## 4. Sclerenchyma — Hardness, Rigidity & Lignin

- Coconut husk ko hard-stiff banane wali tissue 'Sclerenchyma' hai — dead cells, thick lignified walls wali simple permanent tissue hai. *[Q870, Q945, Q974]*
- Sclerenchyma plant ko hard-stiff banata hai — dead cells, highly thickened lignified walls, coconut husk, stems, vascular bundles, leaf veins mein milta hai. *[Q922, Q955]*
- Sclerenchyma cells ki wall 'Lignin' se thick hoti hai — rigidity deta hai, bacterial infections rokta hai. 3 types ground tissue: parenchyma, collenchyma, sclerenchyma. *[Q927, Q932]*

- Sclerenchyma tissue 'dead cells' se bani hoti hai — mechanical stiffness-strength deti hai, fibers-sclereids main cell types hain. Aerenchyma gas exchange, Parenchyma food storage karti hai. *[Q948, Q957, Q965]*
- ⚠ Trap: Sclerenchyma diagrams alag dikh sakte hain — Transverse section (A) mein polygonal cells uniform thickening ke saath, Longitudinal section (B) mein long-narrow-pointed cells dikhte hain. *[Q887]*

## 5. Cork, Cutin & Waterproofing

- Cork water-gases ke liye impermeable hoti hai 'Suberin' ki wajah se — wax-like fatty substance hai cork cell walls mein, water-solutes ke against barrier banata hai. *[Q872]*
- Desert plants ki epidermis par 'Cutin' ka thick waxy coating hota hai — transpiration se water loss prevent karta hai, desert survival ke liye crucial hai. *[Q935, Q1001]*

## 6. Meristematic Tissue & Differentiation

- Meristematic tissue cells division ability lose karke permanent shape-size-function lete hain — is process ko 'Differentiation' kehte hain. Dedifferentiation permanent cells wapas divide karne lagti hain. *[Q873, Q959]*
- 'Meristematic tissue' hi cell division ke capable hoti hai — growing regions mein hoti hai. Types: Apical (length), Lateral (girth), Intercalary (internode size). *[Q934]*
- Meristematic tissue ke cells isolate hokar different permanent tissue banate hain — constant cell division, specialized cells mein differentiate hoti hai. *[Q950]*
- 'Intercalary meristems' leaves ke bottom (base) mein milte hain — nodes-internodes mein elongation promote karte hain, grasses-monocots mein leaf primordia base par. *[Q956]*
- Meristem tissue root tip ya shoot tip mein milti hai — plant growth ke liye responsible hai, actively divide hokar naye cells produce karti hai. *[Q966]*
- Anterior (apical) meristem stem-root ke apex (growing crest) mein milta hai — length badhaane ka function hai. *[Q972, Q973]*
- Lateral meristematic tissue ring-shaped form mein stems mein milti hai, plant ki growth mein help karti hai — girth-thickness badhaati hai (cambium is category ka specific type hai). *[Q986]*

## 7. Fermentation & Anaerobic Respiration — Cytoplasm

- Yeast fermentation aur human muscle cells ki anaerobic respiration mein similarity — dono 'Cytoplasm' mein hoti hain. Yeast ethanol+CO2 banati hai, muscles lactic acid. *[Q874]*
- Yeast fermentation mein Pyruvate 'Ethanol aur Carbon dioxide' mein convert hota hai — glycolysis ke end mein pyruvate banta hai, aerobic respiration CO2+water banati hai. *[Q882]*

## 8. Waste Products in Plants — Gums & Resins

- Tall tree apne waste products 'old xylem (heartwood)' mein store karta hai — gums-resins ki tarah, kyunki purani inner xylem ab water transport nahi karti. *[Q888, Q883]*
- Acacia jaise trees ka gum ek 'waste product' hai — metabolic processes ka byproduct hai, gums-oils-latex-resins old xylem tissues mein store hote hain. *[Q889, Q894]*
- ⚠️ Trap: Plants leaves se 'toxic substances filter' NAHI karte — ye recognized excretion process nahi hai. Plants transpiration, stomata diffusion, old xylem storage, vacuole storage, aur soil mein excretion se waste nikalte hain. *[Q885]*
- Sahi statement: 'Kuch plants mein waste products resins ki tarah store hote hain' — plants soil mein waste excrete karte hain, excess water transpiration se nikalti hai (translocation se NAHI), O2 photosynthesis ka waste product hai. *[Q886]*

## 9. Vascular Tissue — Land Plant Adaptation

- Vascular tissue (Xylem-Phloem) complex plants ki distinctive feature hai — water-minerals-food ka long-distance efficient transport, terrestrial environments mein survival possible banata hai. *[Q884]*

## 10. Xylem — Components & Function

- ⚠️ Trap: 'Xylem fibres' water transport nahi karti — mechanical support deti hain. Xylem vessels-tracheids main water-conducting elements hain, xylem parenchyma starch store karti hai. *[Q891]*
- ⚠️ Trap: 'Sieve tubes' Xylem ka part NAHI hai — Phloem ka part hai. Xylem parts: tracheids, vessels, xylem parenchyma, xylem fibers. *[Q933]*
- Xylem 'roots se materials transport away' karta hai — water-materials upward transport karta hai. *[Q970]*
- ⚠️ Trap: 'Sieve tube ki cell non-living hoti hai' wali baat FALSE hai — companion cells living hoti hain jo sugars sieve elements mein load karne mein help karti hain. Xylem 4 components se banta hai: tracheids, vessels, xylem parenchyma, xylem fibers (ye statement TRUE hai). *[Q996]*
- 'Xylem' example hai permanent tissue ka — meristematic tissue se derive hoti hai, division ability lose kar chuki hai. Blood/Bone/Skin animal tissues hain. *[Q978]*
- Xylem 'complex permanent' tissue hai — vessels, tracheids, parenchyma, fibers se milkar bani hoti hai, water-minerals roots se upward transport karti hai. *[Q991]*

## 11. Seeds — Plumule, Cotyledon & Parts

- 'Plumule' (embryonic shoot) seed mein milta hai — germination par shoot system (stem-leaves) banata hai. Radicle root system banata hai. *[Q892]*
- 'Cotyledons' plant seeds mein food store ki tarah kaam karte hain — embryo ka part hain, germination par pehli leaves ban jaate hain. *[Q926]*
- Seed ke 3 parts: 'Cotyledons, radicle aur Coleoptile' — seed coat aur embryo (radicle+embryonal axis+1-2 cotyledons) se bana hota hai. Coleoptile monocot ki pehli leaf hai. *[Q975]*

## 12. Phloem — Sieve Plates & Food Transport

- Sieve plates par 'Phloem tube tissue' available hoti hai — porous end walls jo sieve tube elements ko continuous sieve tube mein connect karti hain, organic nutrients transport karti hain. *[Q893]*
- Phloem 'photosynthetic parts se material away' transport karti hai — bidirectional vascular tissue hai, organic nutrients leaves se doosre parts tak le jaati hai. *[Q929]*
- Phloem plants mein 'food' ke transport ke liye responsible hai — parenchyma, fibers, sieve tubes, companion cells se bani hoti hai. *[Q931]*
- Phloem 'food ka conduction' karta hai — 5 cell types se bana hai: sieve cells, sieve tubes, companion cells, phloem fibers, phloem parenchyma (fibers chhodkar sabhi living hote hain). *[Q947]*

## 13. Chlamydomonas — Respiration in Aquatic Algae

- Chlamydomonas jaise unicellular algae 'dissolved oxygen ko diffusion se absorb' karte hain (jaise Amoeba) — freshwater-seawater mein rehte hain, specialized respiratory organs nahi hote. *[Q895]*

## 14. Minerals for Pest Resistance

- 'Potassium, Calcium aur Magnesium' crops ko pests withstand karne mein help karte hain — Potassium growth-metabolism, Calcium drought-heat-disease tolerance, Magnesium phosphate metabolism-enzyme activation mein help karte hain. *[Q896]*

## 15. Signs of Life — Movement, Growth & Respiration

- 'Kisi bhi kism ki movement, growth ya respiration' existence ka common proof hai — living organisms ki fundamental characteristics: movement, respiration, sensitivity, growth, reproduction, excretion, nutrition. *[Q897]*

## 16. CAM Photosynthesis — Desert Plants

- 'Desert plants' raat mein CO2 lete hain aur intermediate acid banate hain, jisse din mein chlorophyll energy use hoti hai — CAM (Crassulacean Acid Metabolism), stomata din mein closed rehti hai water loss minimize karne ke liye. *[Q898]*

## 17. Low Energy Needs of Plants

- Plants ki energy needs animals se kam hoti hain kyunki: mobility ki absence hai, aur unka body ka bada hissa dead supportive tissues (xylem-sclerenchyma) se bana hota hai jo metabolic energy maintenance ke liye nahi chahiye. *[Q899]*

## 18. Nitrogen Fixation — Rhizobia Symbiosis

- ⚠ Trap: 'Ficus' rhizobia bacteria ke saath symbiotic relationship NAHI banata, isliye nitrogen fix nahi karta — Lentils, Chickpeas, Peanuts legumes (Fabaceae) hain jo root nodules banakar nitrogen fix karte hain. *[Q900]*

## 19. Angiosperms (Magnoliophyta)

- 'Magnoliophyta' flowering plants hain jinhe 'Angiosperms' bhi kehte hain — seeds ovary (fruit) ke andar enclosed hote hain. Gymnosperms naked seeds, Pteridophytes spores, Bryophytes non-vascular hote hain. *[Q901]*

## 20. Succulent Plants

- ⚠ Trap: Succulent plants mein 'no leaves' universal feature NAHI hai — generally leaves hoti hain (thick-fleshy, water retain karne ke liye). Examples: Aloe Vera, Snake Plant, Jade Plant — drought-resistant, stems-roots present hote hain. *[Q902]*

## 21. Bryophyta — Examples & Characteristics

- Bryophyta (non-vascular land plants — liverworts, hornworts, mosses) ke examples: Marchantia, Riccia, Funaria — Deodar Gymnosperm hai, Marsilea Pteridophyta hai, Cladophora Algae hai. *[Q903, Q946, Q1000]*
- Bryophytes 'plant kingdom ke amphibians' kehlaate hain — soil mein rehte hain lekin sexual reproduction ke liye water par depend karte hain, mosses-liverworts is group mein aate hain. *[Q1006]*
- Rhizoids (Bryophytes ki lower epidermal cells se protuberances) ke baare mein: ⚠ Trap: 'Rhizoids ke fully developed stem-root-leaves hote hain' — ye INCORRECT statement hai, actually rhizoids mein ye developed structures NAHI hote. *[Q977]*
- 'Rhizoids' Bryophytes ki lower epidermal cells se extend hote hain — thin hair-like structures, plant ko anchor karte hain aur water-nutrients absorb karte hain. *[Q980]*
- Group jinke plants roots-stem-leaves nahi rakhte, unhe 'Thallus' kehte hain — Thallus wale plants mein photosynthesis-reproduction-nutrient absorption directly undifferentiated body se hota hai. *[Q976]*

## 22. Medicinal Plants — Sarpagandha & Arjuna

- 'Sarpagandha' (Rauvolfia serpentina) blood pressure treat karta hai — asthma-insomnia bhi treat karta hai, sub-Himalayan forests mein grow karta hai. *[Q904]*
- Arjuna tree ki bark 'Heart Disease' treat karne ke liye use hoti hai — coronary artery disease, heart failure, edema, angina, asthma treat karti hai. *[Q912]*

## 23. Stinging & Carnivorous Plants

- 'Nettle' plant mein stinging hair hote hain jo touch karne par painful stings dete hain — wild mein grow karta hai. *[Q905]*
- ⚠️ Trap: 'Tiger Lily' carnivorous plant NAHI hai — herbaceous perennial hai. Carnivorous plants: Sundew, Corkscrew, Monkeycup, Venus flytrap, Pitcher plant. *[Q906]*

## 24. Thallophyta — Undifferentiated Plant Body

- Spirogyra mein plant body root-stem-leaves mein differentiate NAHI hota — free-floating green algae hai freshwater mein. *[Q907, Q914, Q924]*
- Marsilea mein specialised vascular tissue hoti hai water-substances conduct karne ke liye — Moss/Riccia/Chara mein vascular tissue nahi hoti (Chara green algae hai, plant NAHI). *[Q915]*
- 'Thallophyta' group ko commonly Algae kehte hain — primitive plant life, unicellular se lekar large algae-fungi-lichens tak range karti hai. *[Q940, Q961, Q964]*
- Lichen plant 'Thallophyta' division ka hai — primitive forms, undifferentiated body. *[Q969, Q971, Q967]*
- Plants jinka body differentiated NAHI hota, wo 'Thallophyta' group mein aate hain — non-vascular plants (algae, fungi, lichens), undifferentiated body 'thallus' kehlaata hai. *[Q1004]*

## 25. Potato — An Edible Underground Stem

- Potato ka edible part 'Stem' hai — potatoes special underground stems (tubers) hote hain. Binomial name: Solanum tuberosum. Doosre stem vegetables: onion, ginger, garlic, turmeric. *[Q908]*

## 26. Marine Algae — Pigments

- Marine Algae ka 'Green' color 'Chlorophyll' pigment se hota hai — Phycocyanin cyanobacteria mein blue pigment, Zeaxanthin carotenoid hai, Beta Carotene yellow-orange-red color deta hai. *[Q909]*

## 27. Vascular vs Non-Vascular Plants

- 'Bryophytes' plants hain jinme vascular system nahi hota — Pteridophytes-Tracheophytes vascular tissue rakhte hain (xylem-phloem). *[Q910, Q984]*
- Vessels (water conducting cells) mainly 'Angiosperms' mein milte hain — long cylindrical tube-like structures, lignified walls, large central cavity. *[Q953]*
- 'Gymnosperms' ke paas specialised water-conducting tissue hoti hai — vascular plants hain, ovules ovary wall se enclosed nahi hote (naked seeds). *[Q968]*

## 28. Plant Families

- 'Touch-me-not' (Mimosa pudica) plant 'Mimosaceae' family ka hai — Cyperaceae grass family, Acanthaceae Justicia family, Malvaceae Hibiscus family hai. *[Q911]*

## 29. Petiole — Leaf Stalk

- 'Petiole' ek tree (plant) ka part hai — leaf blade ko leaf base se connect karne wali stalk hai, xylem ke through roots se leaf tak water-nutrients transport karta hai. *[Q913]*

## 30. Plant Kingdom Classification — Groups & Hierarchy

- ⚠ Trap: 'Arthropoda' plant group NAHI hai — ye animal phylum hai (insects, spiders, crustaceans). Plant groups: Thallophyta, Bryophyta, Pteridophyta, Gymnosperms, Angiosperms. *[Q916]*
- Plant classification hierarchy: Kingdom → Division → Class → Order → Family → Genus → Species. *[Q943]*
- 'Gymnosperms' naked seeds produce karte hain (fruit mein enclosed nahi hote) — examples: Pinus, Cycas, Araucaria, Thuja, Cedrus, Picea. *[Q952]*
- 3 types ke simple permanent tissues: 'Parenchyma, Collenchyma aur Sclerenchyma' — Parenchyma food store, Collenchyma flexibility, Sclerenchyma structural hardness deti hai. Phloem complex tissue hai. *[Q982]*
- 'Phanerogams' seed-producing plants hain (gymnosperms + angiosperms) — well-differentiated reproductive tissues rakhte hain jo seeds banate hain. Cryptogams spores se reproduce karte hain. *[Q990]*
- Pteridophyta mein 'flowers' NAHI hote — vascular plants hain jinme true roots-stems-leaves hote hain, spores se reproduce karte hain (fern, horsetail examples). *[Q987, Q989]*
- Cedar (gymnosperm) mein 'uncoated' (naked) seeds milte hain — fruit mein enclosed nahi hote, cones-scales ki surface par expose hote hain. Marsilea Pteridophyta hai (spores se reproduce). *[Q944]*
- 'Gymnosperms' commonly 'exposed' type ke seeds ke liye jaane jaate hain — 'naked seed' ka matlab hai, ovary/fruit mein enclosed nahi hote, cones (strobili) par bante hain. *[Q985]*

## 31. Species — Taxonomic Definition

- Genetically distinct aur reproductively isolated similar organisms ka group 'Species' kehlaata hai — Family related genera ka group hai, Class related orders ka group hai. *[Q917]*

## 31b. Halophytes — Plants Growing in Saline Water

- Saline water mein grow karne wale plants 'Halophytes' kehlaate hain — examples: mangroves, saltwort, pickleweed. Xerophytes dry environments mein, Hydrophytes water mein grow karte hain. *[Q962]*

## 32. Multi-Coloured Leaves — Croton

- 'Croton' (Euphorbiaceae) plant mein multi-coloured leaves hoti hain — multiple pigments (Chlorophyll, Carotenoid, Anthocyanin) alag-alag wavelengths absorb karte hain. *[Q918]*

## 33. Fungi & Algae — Growth Without Sunlight

- Fungi paani mein sunlight ki absence mein grow kar sakte hain — eukaryotic organisms (yeasts, moulds, mushrooms) hain, Algae photosynthetic organisms hain. *[Q919]*

## 34. Bacterial Flagella Arrangements

- Bacterium mein flagella dono ends par ho to use 'Amphitrichous' kehte hain — Peritrichous flagella pure body mein, Monotrichous single polar flagellum hota hai. *[Q920]*

## 35. Collenchyma — Flexibility

- Plants mein flexibility 'Collenchyma' tissue se hoti hai — bending allow karta hai bina toote, tendrils-climbers mein hota hai. Living cells, corners par irregularly thickened. *[Q921, Q988, Q995]*

## 36. Parenchyma — Loosely Packed Cells

- 'Parenchyma' cells loosely packed hote hain, large intercellular spaces hote hain — food store karti hai (chlorenchyma-aerenchyma isi mein aate hain). *[Q923, Q937, Q954, Q960]*
- Parenchyma tissue 'relatively unspecialised cells with thin walls' se bani hoti hai — cellulose se bane thin cell walls, living cells, ground tissue ka bulk banati hai. *[Q938]*

## 37. Root Hair

- Root cells ke long hair-like parts 'Root hair' kehlaate hain — fine, thread-like structures, epidermal cells se project hote hain, surface area badhaate hain water-minerals absorption ke liye. *[Q930]*

## 38. Dicot vs Monocot — Roots, Cotyledons & Venation

- ⚠️ Trap: 'Fibrous root' dicotyledonous plants ki property NAHI hai — monocots ki characteristic hai (maize, grass, wheat). Dicots (gram, neem) tap root, 2 cotyledons, reticulate venation rakhte hain. *[Q939]*

- Kalmi (Water spinach) jaise plants ke seeds mein 'two cotyledons' hote hain (dicotyledon) — apples, mangoes, peas doosre examples. Ulva algae, Cedar-Pine gymnosperms hain. *[Q942]*
- 'Garlic' ek monocot plant hai — sirf ek cotyledon hota hai. Doosre monocots: sugarcane, banana, daffodils, palm. *[Q949]*
- 'Dicotyledon plants' angiosperms hain jinke seeds mein 2 cotyledons hote hain — Monocots (maize, grass) mein sirf 1 cotyledon hota hai. *[Q981]*
- 'Paphiopedilum' (Venus slipper orchid) ek monocot hai — Orchidaceae family monocots ki hai. Monocots mein 1 seed leaf (cotyledon) hota hai. *[Q994]*

## 39. Rhizomes — Underground Stems

- 'Ginger' Rhizome ka classic example hai — horizontal underground stem, shoot-root systems produce karta hai. Doosre examples: bamboo, turmeric, lotus, asparagus. *[Q941]*

## 40. Transpiration — Excess Water Removal

- Plants excess water 'Transpiration' se remove karte hain — leaf surface (stomata) se water vapour evaporate hota hai. *[Q936]*

## 41. Transpiration — Advantages & Disadvantages

- Transpiration ke advantages: cooling effect, excess water removal, mineral transport, turgidity maintain, photosynthesis ke liye water supply — ⚠️ Disadvantage: rate absorption se zyada hone par 'wilting' ho sakti hai. *[Q881]*
- Roots plants ko water-mineral ions transport karne mein: pehle 'Ions active transport se enter karte hain', phir 'Water concentration gradient ke along follow karta hai' — energy use hoti hai ion transport mein. *[Q880]*

## 42. Plant Roots — Oxygen Absorption

- Plant roots 'soil particles ke beech ki air spaces' se oxygen lete hain — respiration ke liye zaroori hai. Root types: Tap Root, Fibrous Root — Functions: Absorption, Anchoring, Storage. *[Q879]*

## 43. Monohybrid Cross

- Monohybrid cross 2 plants ko combine karta hai jinke trait ke 'do' (two) different versions hote hain — homozygous genotypes wale individuals ka genetic mix hota hai. *[Q878]*

## 44. Simple vs Complex Permanent Tissue

- ⚠ Trap: 'Xylem' simple permanent tissue NAHI hai — Xylem-Phloem complex permanent tissues hain (multiple cell types se bani). Sclerenchyma, Parenchyma, Collenchyma simple permanent tissues hain. *[Q925]*
- 'Xylem' complex permanent tissue hai — sirf ek cell type se nahi bani (tracheids, vessels, xylem parenchyma, xylem fibers ka mix hai). Phloem bhi complex hai. *[Q928, Q993]*
- 'Complex tissue' alag-alag cell types se milkar same function perform karti hai (jaise xylem-phloem) — Meristematic tissue constantly divide karti hai, Epidermal tissue protective covering hai. *[Q963]*
- 'Parenchyma aur Collenchyma' — dono living cells hoti hain. Sclerenchyma dead cells se bani permanent tissue hai jo plant ko hard-stiff banati hai. *[Q951]*

## 45. Skin/Epidermal Cells

- Skin (aerial plant parts) 'Epidermal cells' se bani hoti hai — waxy water-resistant layer secrete karti hai, protection deti hai water loss-mechanical injury-fungi se. *[Q958]*

## 46. Cuscuta — A Parasitic Plant

- 'Cuscuta' (Dodder/Amarbel) parasitic plant hai — chlorophyll-leaves nahi hoti, yellow-coloured climber hai, haustoria banakar host plant ke phloem se nutrition leta hai. *[Q997]*

## 47. Vegetative Parts of a Plant

- Plant ke vegetative parts hain 'Roots, stems aur leaves' — directly sexual reproduction mein involve nahi hote, asexual reproduction (cuttings, budding, grafting) mein use hote hain. *[Q998]*

## 48. Cytokinins — Plant Hormone

- 'Cytokinins' plants mein cell division promote karte hain — roots-shoot systems mein, naye leaves-chloroplasts formation, lateral shoot growth mein bhi help karte hain. *[Q999]*

## 49. Vascular Tissue — Overall Transport in Trees

- Trees mein materials ki transportation 'Vascular tissue' (Xylem+Phloem) se hoti hai — Xylem water-nutrients upward, Phloem food doosre parts tak transport karta hai. *[Q1002]*

## 50. Bryophyllum — Buds on Leaf Margins

- 'Bryophyllum' (Crassulaceae family) ki leaf margins ke notches mein buds produce hote hain — vegetative propagation ke liye plantlet buds banata hai. Rose stem cutting se propagate hota hai. *[Q1005]*

## 51. Fragmentation — Spirogyra

- 'Spirogyra' (Water Silk) green algae fragmentation se asexually reproduce karta hai — Yeast-Hydra budding se reproduce karte hain. *[Q1007]*